

main.py

1# WRITE YOUR CODE HERE

Terminal

Welcome to Ubuntu 18.04.5 LTS (GNU/Linux 5.19.0-1028-aws x86\_64)  
  
\* Documentation: <https://help.ubuntu.com>  
\* Management: <https://landscape.canonical.com>  
\* Support: <https://ubuntu.com/advantage>  
  
\* Canonical Livepatch is available for installation.  
- Reduce system reboots and improve kernel security. Activate at:  
<https://ubuntu.com/livepatch>  
\*  
\* Welcome to the Codio Terminal!  
\*  
\* <https://docs.codio.com/develop/develop/ide/boxes/overview>  
\*  
\* Your Codio Box domain is: cosmoscamera-savagemelody.codio.io  
\*  
Last login: Fri Jul 21 22:43:30 2023 from 192.168.11.179  
codio@cosmoscamera-savagemelody:~/workspace\$ python3 main.py 13  
codio@cosmoscamera-savagemelody:~/workspace\$ python3 main.py 14  
codio@cosmoscamera-savagemelody:~/workspace\$

Guide

Collapse

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Filters

```
print( "+Add a filter to your image ,
```

TRY IT!

Then we are going to create an extra option after option 3. Let's call it option 4.

```
elif option=="4":  
    image_path = input("Enter the path of the image you want to add fi  
    # run filter function  
    filters(image_path)
```

Now we are ready to add our extra function that adds filters using PIL. Our function should ask the user to pick which filter they want to apply. The options are :

1. `contour`: The contour filter traces the edges of objects in the image, creating a high-contrast, black-and-white image that highlights the shapes in the scene.

2. `EDGE_ENHANCE`: The edge enhance filter highlights the edges of objects in the image, making them more distinct and pronounced. This filter can be used to create a more stylized or artistic effect.

3. `FIND_EDGES`: The find edges filter detects the edges of objects in the image and creates a high-contrast, black-and-white image that highlights these edges.

First, our function should open an image file. Then we prompt the user to choose one of the three provided filters to apply to the image. If the user's input is not among the three options (1, 2, or 3), we default to the Contour filter. After applying the chosen filter, we save the result to `filtered_image.jpg` in the current directory.

```
def filters(image_path):  
    # Open an image file  
    with Image.open(image_path) as img:  
        # Ask the user to choose a filter  
        print("Choose a filter to apply on the image:")  
        print("1. Contour")  
        print("2. Edge Enhance")  
        print("3. Find Edges")  
        option = input("Enter your option: ")  
  
        if option == "1":  
            filtered_img = img.filter(ImageFilter.CONTOUR)  
        elif option == "2":  
            filtered_img = img.filter(ImageFilter.EDGE_ENHANCE)  
        elif option == "3":  
            filtered_img = img.filter(ImageFilter.FIND_EDGES)  
        else:  
            print("Invalid option. Applying Contour filter by default.")  
            filtered_img = img.filter(ImageFilter.CONTOUR)  
  
        # Save the filtered image  
        filtered_img.save('filtered_image.png')  
        print("Filter applied successfully, 'filtered_image.png' created.")
```

TRY IT!